

MA1002 — Calculus for Science Part 2

Summer 2024 Examination · Paper 1 · Full Marking Scheme

Module	MA1002 — Calculus for Science Part 2
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Institution	University College Cork
Session	Summer 2024, Paper 1
Duration	1.5 hours (+ 15 minutes reading time)
Structure	Answer any 4 of the 5 questions . Each question carries equal marks.
Materials	Formula sheet supplied. Show all work and say where a calculator has been used.

How to use this scheme. Every answer is worked out in small numbered steps, with a “Rules you need” box listing the rules used. Nothing is skipped, so you can see where every number comes from. Read the answer first, then follow the steps.

Coverage note. On the day you answer 4 questions, but this scheme works through **all five** in full, so you can pick the ones you’re strongest on.

Source basis. Every answer uses only MA1002 course methods: indefinite integrals, substitution, integration by parts, definite integrals and area, volumes of rotation, partial fractions and separable differential equations. Q5(a) (line of best fit) isn’t a listed course topic; the question supplies its formulas, and a note marks it.

Mark splits. The paper doesn’t print marks for each part. This scheme gives each question 25 marks. The split between parts and within each part is Study-Uni’s suggested allocation.

Own-figure rule. Where a later step uses an earlier result, a correct method applied to a wrong earlier value still earns the method marks.

Rounding. Answers are given exactly (e.g. $(\pi/2)(e^4 - 1)$), with a decimal to 2 decimal places where useful. Q5(a) is given to 3 decimal places as the question asks.

Question 1

Antiderivatives (the indefinite integral), initial conditions — 25 marks

1(a)

[12 marks]

Find all $g(x)$ such that $g'(x) = 3e^{4x} - 2x^4 + 4\sqrt{x} - \frac{2}{x} + 6$.

ANSWER

$$g(x) = \frac{3}{4}e^{4x} - \frac{2}{5}x^5 + \frac{8}{3}x^{3/2} - 2\ln|x| + 6x + C$$

RULES YOU NEED

- $\int e^{ax} dx = (1/a) e^{ax}$
- $\int x^n dx = x^{n+1}/(n+1)$ ($n \neq -1$), and $\int (1/x) dx = \ln|x|$
- $\sqrt{x} = x^{1/2}$
- "All $g(x)$ " means add $+ C$.

WORKING — STEP BY STEP

1 Rewrite the root as a power

The power rule only works on powers, so write $\sqrt{x} = x^{1/2}$:

$$g'(x) = 3e^{4x} - 2x^4 + 4x^{1/2} - 2 \cdot \frac{1}{x} + 6$$

2 Integrate the exponential

Here $a = 4$, so divide by 4:

$$3e^{4x} \rightarrow 3 \times \frac{1}{4}e^{4x} = \frac{3}{4}e^{4x}$$

3 Integrate the powers of x

Add 1 to the power, then divide by the new power:

$$-2x^4 \rightarrow -2 \cdot \frac{x^5}{5} = -\frac{2}{5}x^5$$

$$4x^{1/2} \rightarrow 4 \cdot \frac{x^{3/2}}{3/2} = 4 \times \frac{2}{3}x^{3/2} = \frac{8}{3}x^{3/2}$$

($1/2 + 1 = 3/2$, and dividing by $3/2$ is the same as multiplying by $2/3$.)

4 Integrate $2/x$ and the constant

$$-2 \cdot (1/x) \rightarrow -2 \ln|x| \quad 6 \rightarrow 6x$$

5 Put it together, + C

$$g(x) = \frac{3}{4}e^{4x} - \frac{2}{5}x^5 + \frac{8}{3}x^{3/2} - 2 \ln|x| + 6x + C$$

EXAM TECHNIQUE

$2/x$ is not a power-rule term: x^{-1} would give $x^0/0$. It integrates to $2 \ln|x|$. And don't forget the + C when asked for "all" functions.

Marks	Awarded for
2	$3e^{4x} \rightarrow (3/4)e^{4x}$.
2	$-2x^4 \rightarrow -(2/5)x^5$.
3	$4\sqrt{x} \rightarrow (8/3)x^{3/2}$.
2	$-2/x \rightarrow -2 \ln x $.
1	$6 \rightarrow 6x$.
2	+ C included.

1(b)**[13 marks]**Find $f(x)$ if $f''(x) = 2x^2 - 3x + 4$ given that $f(1) = 5$, $f'(1) = 6$.**ANSWER**

$$f(x) = \frac{1}{6}x^4 - \frac{1}{2}x^3 + 2x^2 + \frac{17}{6}x + \frac{1}{2}$$

RULES YOU NEED

- $\int x^n dx = x^{n+1}/(n+1)$.
- Integrate f'' to get f' (constant C), then f' to get f (constant D).
- $f'(1) = 6$ finds C; $f(1) = 5$ finds D.

WORKING — STEP BY STEP**1 Integrate $f''(x)$ to get $f'(x)$**

$$2x^2 \rightarrow \frac{2x^3}{3} \quad -3x \rightarrow -\frac{3x^2}{2} \quad 4 \rightarrow 4x$$

$$f'(x) = \frac{2}{3}x^3 - \frac{3}{2}x^2 + 4x + C$$

2 Use $f'(1) = 6$ to find CPut $x = 1$ in. Use sixths to add the fractions:

$$\frac{2}{3} - \frac{3}{2} + 4 + C = 6$$

$$\frac{4}{6} - \frac{9}{6} + \frac{24}{6} = \frac{19}{6} \Rightarrow \frac{19}{6} + C = 6$$

$$C = \frac{36}{6} - \frac{19}{6} = \frac{17}{6}$$

3 Integrate $f'(x)$ to get $f(x)$

$$\frac{2}{3}x^3 \rightarrow \frac{2}{3} \cdot \frac{x^4}{4} = \frac{1}{6}x^4 \quad -\frac{3}{2}x^2 \rightarrow -\frac{3}{2} \cdot \frac{x^3}{3} = -\frac{1}{2}x^3$$

$$4x \rightarrow 2x^2 \quad \frac{17}{6} \rightarrow \frac{17}{6}x$$

$$f(x) = \frac{1}{6}x^4 - \frac{1}{2}x^3 + 2x^2 + \frac{17}{6}x + D$$

4 Use $f(1) = 5$ to find DPut $x = 1$ in, again in sixths:

$$\frac{1}{6} - \frac{3}{6} + \frac{12}{6} + \frac{17}{6} + D = 5$$

$$\frac{27}{6} + D = 5 \Rightarrow D = \frac{30}{6} - \frac{27}{6} = \frac{3}{6} = \frac{1}{2}$$

5 Check

$$f(1) = \frac{1}{6} - \frac{1}{2} + 2 + \frac{17}{6} + \frac{1}{2} = 5 \quad \checkmark \quad f'(1) = \frac{2}{3} - \frac{3}{2} + 4 + \frac{17}{6} = 6 \quad \checkmark$$

EXAM TECHNIQUE

The conditions are at $x = 1$, not $x = 0$, so the constants don't just drop out. Put every term over the same denominator (6) to avoid fraction slips.

Marks	Awarded for
3	$f'(x) = (2/3)x^3 - (3/2)x^2 + 4x + C$.
3	$C = 17/6$ from $f'(1) = 6$.
3	Integrating again: $(1/6)x^4 - (1/2)x^3 + 2x^2 + (17/6)x + D$.
3	$D = 1/2$ from $f(1) = 5$.
1	Final $f(x)$ stated.

Question 1 Total: 25 Marks

Question 2

Integration by parts, integration by substitution — 25 marks

2(a)

[15 marks]

Use integration by parts to evaluate $\int (t^3 - 2t^2 + t) \ln t \, dt$.

ANSWER

$$\left(\frac{t^4}{4} - \frac{2t^3}{3} + \frac{t^2}{2}\right) \ln t - \frac{t^4}{16} + \frac{2t^3}{9} - \frac{t^2}{4} + C$$

RULES YOU NEED

- Integration by parts: $\int u \, v' \, dt = u \, v - \int u' \, v \, dt$.
- When $\ln t$ is in the integral, let $u = \ln t$ (we can differentiate it, but not easily integrate it).
- $d/dt (\ln t) = 1/t$.

WORKING — STEP BY STEP

1 Choose u and v'

Let $u = \ln t$ and $v' =$ the polynomial:

$$u = \ln t \Rightarrow u' = \frac{1}{t}$$

$$v' = t^3 - 2t^2 + t \Rightarrow v = \frac{t^4}{4} - \frac{2t^3}{3} + \frac{t^2}{2}$$

2 Apply the parts formula

$$\int (t^3 - 2t^2 + t) \ln t \, dt = \left(\frac{t^4}{4} - \frac{2t^3}{3} + \frac{t^2}{2}\right) \ln t - \int \frac{1}{t} \left(\frac{t^4}{4} - \frac{2t^3}{3} + \frac{t^2}{2}\right) dt$$

3 Simplify $u'v$

Divide each term by t (take 1 off each power):

$$\frac{1}{t} \left(\frac{t^4}{4} - \frac{2t^3}{3} + \frac{t^2}{2}\right) = \frac{t^3}{4} - \frac{2t^2}{3} + \frac{t}{2}$$

4 Integrate it

$$\frac{t^3}{4} \rightarrow \frac{t^4}{16} \quad -\frac{2t^2}{3} \rightarrow -\frac{2t^3}{9} \quad \frac{t}{2} \rightarrow \frac{t^2}{4}$$

5 Put it together

Subtract the whole bracket, so every sign flips:

$$\left(\frac{t^4}{4} - \frac{2t^3}{3} + \frac{t^2}{2}\right) \ln t - \left(\frac{t^4}{16} - \frac{2t^3}{9} + \frac{t^2}{4}\right) + C$$

$$= \left(\frac{t^4}{4} - \frac{2t^3}{3} + \frac{t^2}{2}\right) \ln t - \frac{t^4}{16} + \frac{2t^3}{9} - \frac{t^2}{4} + C$$

EXAM TECHNIQUE

Don't pick $u = t^3 - 2t^2 + t$: then you'd need $v = \int \ln t \, dt$, which is harder. With $\ln t$ present, $u = \ln t$ almost always.

Marks	Awarded for
3	$u = \ln t$, $u' = 1/t$.
3	$v = t^4/4 - 2t^3/3 + t^2/2$.
3	Parts formula applied correctly.
2	$u'v$ simplified to $t^3/4 - 2t^2/3 + t/2$.
2	Integrated: $t^4/16 - 2t^3/9 + t^2/4$.
2	Final answer with signs correct, $+ C$.

2(b)**[10 marks]**

Use integration by substitution to evaluate $\int x^2(x^3 - 3)^{2024} dx$.

ANSWER

$$\frac{(x^3 - 3)^{2025}}{6075} + C$$

RULES YOU NEED

- Substitution: let u = the inside of the bracket; $du = (du/dx) dx$.
- $\int u^n du = u^{n+1}/(n + 1)$.

WORKING — STEP BY STEP**1 Choose u**

Take the inside of the bracket. Its derivative, $3x^2$, matches the x^2 outside:

$$u = x^3 - 3$$

2 Find du and make $x^2 dx$ on its own

$$\frac{du}{dx} = 3x^2 \Rightarrow du = 3x^2 dx \Rightarrow x^2 dx = \frac{1}{3} du$$

3 Substitute

$$\int u^{2024} \cdot \frac{1}{3} du = \frac{1}{3} \int u^{2024} du$$

4 Integrate with the power rule

$2024 + 1 = 2025$, and $3 \times 2025 = 6075$:

$$\frac{1}{3} \cdot \frac{u^{2025}}{2025} + C = \frac{u^{2025}}{6075} + C$$

5 Put x back in

$$\frac{(x^3 - 3)^{2025}}{6075} + C$$

EXAM TECHNIQUE

Never try to expand $(x^3 - 3)^{2024}$. A huge power on a bracket is the signal to substitute for the bracket.

Marks	Awarded for
3	$u = x^3 - 3, du = 3x^2 dx.$
2	Integral rewritten as $(1/3) \int u^{2024} du.$
3	Integrated: $u^{2025}/6075.$
2	Back in terms of $x, + C.$

Question 2 Total: 25 Marks

Question 3

Definite integrals by substitution, area between two curves — 25 marks

3(a)

[12 marks]

Evaluate $\int_1^5 \sqrt{2x+5} \, dx$.

ANSWER

$$\frac{15\sqrt{15} - 7\sqrt{7}}{3} \approx 13.19$$

RULES YOU NEED

- Substitution $u = 2x + 5$, $du = 2 \, dx$. Change the limits to u -values.
- $\int u^{1/2} \, du = u^{3/2}/(3/2) = (2/3)u^{3/2}$.
- $u^{3/2} = u\sqrt{u}$.

WORKING — STEP BY STEP

1 Substitute

Let u be the inside of the root:

$$u = 2x + 5 \Rightarrow du = 2 \, dx \Rightarrow dx = \frac{1}{2} \, du$$

2 Change the limits

$$x = 1 \Rightarrow u = 2(1) + 5 = 7 \quad x = 5 \Rightarrow u = 2(5) + 5 = 15$$

3 Rewrite and integrate

$$\int_7^{15} u^{1/2} \cdot \frac{1}{2} \, du = \frac{1}{2} \left[\frac{2}{3} u^{3/2} \right]_7^{15} = \frac{1}{3} \left[u^{3/2} \right]_7^{15}$$

4 Put in the limits

$$\frac{1}{3} (15^{3/2} - 7^{3/2}) = \frac{1}{3} (15\sqrt{15} - 7\sqrt{7})$$

5 Decimal (calculator)

$$15\sqrt{15} \approx 58.095 \quad 7\sqrt{7} \approx 18.520 \Rightarrow \frac{58.095 - 18.520}{3} = \frac{39.575}{3} \approx 13.19$$

EXAM TECHNIQUE

The paper says to show where a calculator was used, so write “(calculator)” beside the decimal step. Keep the exact form too.

Marks	Awarded for
3	$u = 2x + 5, dx = (1/2) du.$
2	New limits 7 and 15.
3	Integrated: $(1/3)u^{3/2}.$
3	$(1/3)(15\sqrt{15} - 7\sqrt{7}).$
1	$\approx 13.19.$

3(b)**[13 marks]**

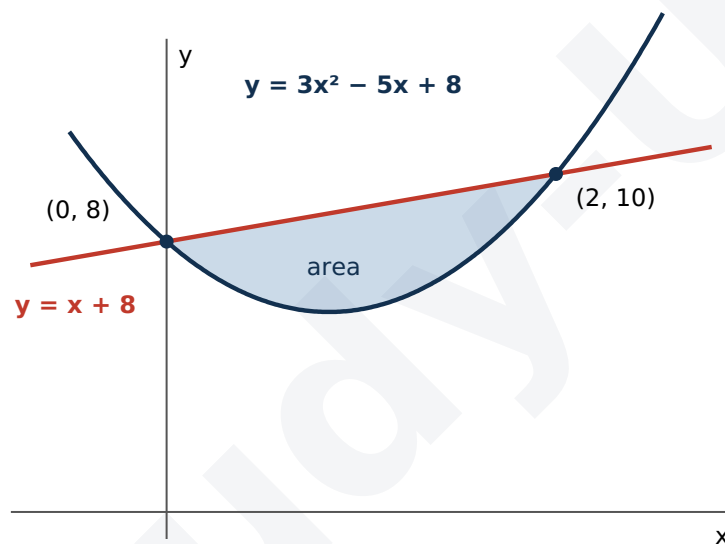
Compute the area of the region bound by the line $y = x + 8$ and the parabola $y = 3x^2 - 5x + 8$.

ANSWER

Area = 4 square units

RULES YOU NEED

- Area between two curves = $\int (\text{upper curve} - \text{lower curve}) dx$.
- The limits are the x-values where the curves meet.



Between $x = 0$ and $x = 2$ the line is above the parabola. The shaded region is the area we want.

WORKING — STEP BY STEP**1 Find where they meet**

Set them equal, bring everything to one side and factor:

$$x + 8 = 3x^2 - 5x + 8 \Rightarrow 0 = 3x^2 - 6x \Rightarrow 3x(x - 2) = 0$$

$$x = 0 \quad \text{or} \quad x = 2$$

2 Which is on top?

Test $x = 1$:

$$\text{Line: } 1 + 8 = 9 \quad \text{Parabola: } 3 - 5 + 8 = 6 \Rightarrow \text{line on top}$$

3 Set up (upper – lower)

The 8s cancel:

$$A = \int_0^2 [(x + 8) - (3x^2 - 5x + 8)] dx = \int_0^2 (6x - 3x^2) dx$$

4 Integrate

$6x \rightarrow 3x^2$, and $3x^2 \rightarrow x^3$:

$$A = [3x^2 - x^3]_0^2$$

5 Put in the limits

$$x = 2: 3(4) - 8 = 12 - 8 = 4 \quad x = 0: 0$$

$$A = 4 - 0 = 4$$

EXAM TECHNIQUE

Bracket the whole parabola when subtracting: $-(3x^2 - 5x + 8) = -3x^2 + 5x - 8$. Missing the bracket is the most common slip here.

Marks	Awarded for
3	Intersections $x = 0$ and $x = 2$.
2	Line identified as the upper curve.
3	Integral set up: $\int_0^2 (6x - 3x^2) dx$.
3	Integrated: $3x^2 - x^3$.
2	Area = 4.

Question 3 Total: 25 Marks

Question 4

Volumes of rotation, partial fractions — 25 marks

4(a)

[12 marks]

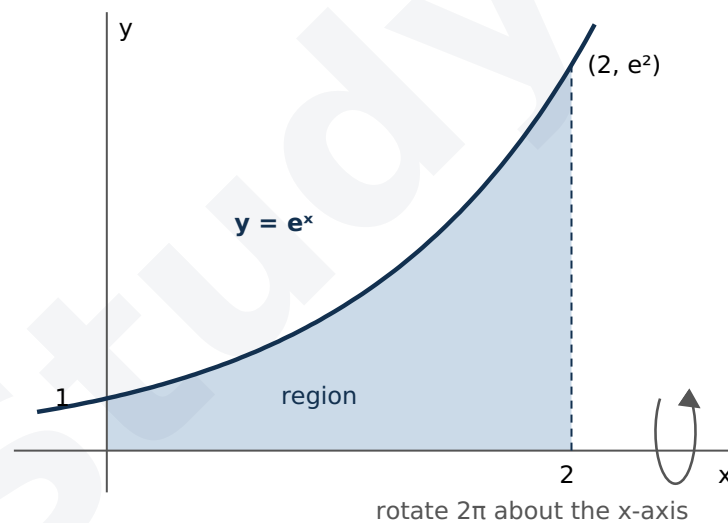
Compute the volume obtained by rotating through 2π radians about the x axis the region in the x - y plane bound by $x = 0$, $y = e^x$, the x -axis, and $x = 2$.

ANSWER

$$V = \frac{\pi}{2}(e^4 - 1) \approx 84.19 \text{ cubic units}$$

RULES YOU NEED

- Volume of revolution about the x -axis: $V = \int_a^b \pi y^2 dx$.
- $(e^x)^2 = e^{2x}$, and $\int e^{2x} dx = (1/2)e^{2x}$.
- $e^0 = 1$.



The region under $y = e^x$ from $x = 0$ to $x = 2$ spins a full turn (2π) around the x -axis.

WORKING — STEP BY STEP

1 Square y

$$y^2 = (e^x)^2 = e^{2x}$$

2 Set up the volume

Limits are $x = 0$ and $x = 2$:

$$V = \int_0^2 \pi e^{2x} dx = \pi \int_0^2 e^{2x} dx$$

3 IntegrateHere $a = 2$:

$$V = \pi \left[\frac{1}{2} e^{2x} \right]_0^2$$

4 Put in the limits

$$\pi \left(\frac{1}{2} e^4 - \frac{1}{2} e^0 \right) = \pi \left(\frac{1}{2} e^4 - \frac{1}{2} \right) = \frac{\pi}{2} (e^4 - 1)$$

5 Decimal (calculator)

$$e^4 \approx 54.598 \Rightarrow \frac{\pi}{2} \times 53.598 \approx 84.19$$

EXAM TECHNIQUE

Square y before integrating: $(e^x)^2 = e^{2x}$, not e^{x^2} . The lower limit gives $e^0 = 1$, not 0.

Marks	Awarded for
3	$y^2 = e^{2x}$.
2	$V = \pi \int_0^2 e^{2x} dx$ set up.
3	Integrated: $(1/2)e^{2x}$.
3	$(\pi/2)(e^4 - 1)$.
1	≈ 84.19 .

4(b)**[13 marks]**

Use the method of partial fractions to compute $\int \frac{12x - 14}{x^2 - 2x - 15} dx$.

ANSWER

$$\frac{23}{4} \ln|x - 5| + \frac{25}{4} \ln|x + 3| + C$$

RULES YOU NEED

- Factor the bottom, then write the fraction as $A/(\text{first factor}) + B/(\text{second factor})$.
- Find A and B by substituting the x-values that make each bracket zero.
- $\int 1/(x - a) dx = \ln|x - a|$.

WORKING — STEP BY STEP**1 Factor the denominator**

Two numbers that multiply to -15 and add to -2 : -5 and $+3$.

$$x^2 - 2x - 15 = (x - 5)(x + 3)$$

2 Set up the partial fractions

Multiply both sides by $(x - 5)(x + 3)$:

$$\frac{12x - 14}{(x - 5)(x + 3)} = \frac{A}{x - 5} + \frac{B}{x + 3} \Rightarrow 12x - 14 = A(x + 3) + B(x - 5)$$

3 Find A: put $x = 5$

The B bracket becomes 0:

$$12(5) - 14 = A(8) \Rightarrow 46 = 8A \Rightarrow A = \frac{46}{8} = \frac{23}{4}$$

4 Find B: put $x = -3$

The A bracket becomes 0:

$$12(-3) - 14 = B(-8) \Rightarrow -50 = -8B \Rightarrow B = \frac{50}{8} = \frac{25}{4}$$

Quick check: $A + B = 48/4 = 12$, the coefficient of x on top ✓.

5 Integrate each fraction

$$\int \left(\frac{23/4}{x - 5} + \frac{25/4}{x + 3} \right) dx = \frac{23}{4} \ln|x - 5| + \frac{25}{4} \ln|x + 3| + C$$

EXAM TECHNIQUE

Watch the signs when you substitute $x = -3$: $12(-3) - 14 = -36 - 14 = -50$, and $(-3) - 5 = -8$.

Marks	Awarded for
3	Denominator factored: $(x - 5)(x + 3)$.
2	$12x - 14 = A(x + 3) + B(x - 5)$ set up.
2.5	$A = 23/4$.
2.5	$B = 25/4$.
3	Integrated to logs, + C.

Question 4 Total: 25 Marks

Question 5

Line of best fit, separable differential equations — 25 marks

5(a)

[12 marks]

Recall, for a collection of n points given by (x_i, y_i) , the line of best fit is a line of the form $y = mx + c$, with $m = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{n\sum x_i^2 - (\sum x_i)^2}$, $c = \frac{\sum y_i - m\sum x_i}{n}$. Use these formulas to compute the line of best fit for the points below, with m and c accurate to 3 decimal places.

x : 4, 11, 19, 24 y : 22, 20, 13, 4

ANSWER

$$m \approx -0.878, \quad c \approx 27.476 \Rightarrow y = -0.878x + 27.476$$

NOTE

Line of best fit isn't one of the listed MA1002 course topics. The question gives you both formulas, so it only needs careful arithmetic.

RULES YOU NEED

- Make a table with rows for x , y , xy and x^2 , then add up each row.
- $\sum x^2$ means square each x , then add. $(\sum x)^2$ means add the x 's, then square.
- Use the unrounded m when working out c . Round only at the very end.

WORKING — STEP BY STEP

1 Build the table of sums

Here $n = 4$ (four points):

	1	2	3	4	Sum (Σ)
x	4	11	19	24	58
y	22	20	13	4	59
xy	88	220	247	96	651
x²	16	121	361	576	1074

2 Top of the m formula

$$n\sum xy - \sum x \sum y = 4 \times 651 - 58 \times 59 = 2604 - 3422 = -818$$

3 Bottom of the m formula

$$n\Sigma x^2 - (\Sigma x)^2 = 4 \times 1074 - 58^2 = 4296 - 3364 = 932$$

4 Find m (calculator)

$$m = \frac{-818}{932} = -0.877682... \approx -0.878$$

5 Find c using the full m (calculator)

$-m \times 58$ is + because m is negative:

$$m\Sigma x = -0.877682... \times 58 = -50.905579...$$

$$c = \frac{59 - (-50.905579...)}{4} = \frac{109.905579...}{4} = 27.476395... \approx 27.476$$

6 Write the line

$$y = -0.878x + 27.476$$

EXAM TECHNIQUE

Rounding m to -0.878 first gives $c = (59 + 50.924)/4 = 27.481$, which is wrong to 3 decimal places. Keep m in your calculator's memory.

Marks	Awarded for
4	Sums $\Sigma x = 58$, $\Sigma y = 59$, $\Sigma xy = 651$, $\Sigma x^2 = 1074$ (1 each).
3	$m = -818/932 \approx -0.878$.
3	$c \approx 27.476$ using the unrounded m.
2	Line $y = -0.878x + 27.476$ stated.

5(b)**[13 marks]**

Solve the following ordinary differential equation using the method of separation of variables: $\frac{dy}{dx} = e^{-y}(2x - 4)$ given that $y = 0$ when $x = 2$.

ANSWER

$$y = \ln(x^2 - 4x + 5)$$

RULES YOU NEED

- Separation of variables: all y terms with dy on one side, all x terms with dx on the other, then integrate both sides.
- $1/e^{-y} = e^y$, and $\int e^y dy = e^y$.
- If $e^y = k$, then $y = \ln k$. Also $e^0 = 1$.

WORKING — STEP BY STEP**1 Separate the variables**

Divide both sides by e^{-y} (the same as multiplying by e^y), then multiply by dx :

$$e^y dy = (2x - 4) dx$$

2 Integrate both sides

$$\int e^y dy = \int (2x - 4) dx \Rightarrow e^y = x^2 - 4x + C$$

3 Use $y = 0$ when $x = 2$

$$e^0 = 2^2 - 4(2) + C \Rightarrow 1 = 4 - 8 + C \Rightarrow C = 5$$

$$e^y = x^2 - 4x + 5$$

4 Solve for y

Take $\ln$ of both sides:

$$y = \ln(x^2 - 4x + 5)$$

This works for every x , since $x^2 - 4x + 5 = (x - 2)^2 + 1$ is always positive.

5 Check

$$x = 2: y = \ln(4 - 8 + 5) = \ln 1 = 0 \checkmark$$

EXAM TECHNIQUE

Find C while the equation is still $e^y = x^2 - 4x + C$, before taking logs. Writing $y = \ln(x^2 - 4x) + C$ is wrong: the constant must go inside the log.

Marks	Awarded for
3	Separated: $e^y dy = (2x - 4) dx$.
2	$\int e^y dy = e^y$.
2	$\int (2x - 4) dx = x^2 - 4x + C$.
3	$C = 5$ from $y(2) = 0$.
3	$y = \ln(x^2 - 4x + 5)$.

Question 5 Total: 25 Marks